GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form A · 24 points · two class periods

Name_____
Date_____
Period_____

This assessment runs over two class periods. Stop when the period ends; you will continue from where you stopped.

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors

10 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 4 cm side of a figure is 10 cm in its scale copy. Find the scale factor. *1 pt*

2. A 15 cm side of a figure is 6 cm in its scale copy. Find the scale factor. *1 pt*

3. A figure is scaled by a scale factor of 3. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each side is 3 times as long.
- ☐ B. Each angle is 3 times as large.
- ☐ C. The perimeter is 3 times as long.
- ☐ D. The area is 3 times as large.
- ☐ E. The area is 9 times as large.

4. A 3 cm by 4 cm rectangle is scaled by k = 3.

a. Find the copy's perimeter. *1 pt*

b. Find the copy's area. *1 pt*

5. A figure has area 8 cm^2. Its scale copy has area 72 cm^2. What is the scale factor? *1 pt*

6. A floor plan is drawn at ¼ in : 1 ft. A room is 3 in by 3.5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

b. Find the room's real area. *1 pt*

7. A plan is drawn at 1 in : 4 ft. A real wall is 18 ft long. How long is it on the plan? *1 pt*

8. A plan is drawn at 1 in : 5 ft. A room is 2 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

Part 2. Converting Across Systems *8 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 9 in? *1 pt*

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 70 cm. *1 pt*

- ☐ A. 70 × 0.39
- ☐ B. 70 × 2.54
- ☐ C. 70 ÷ 2.54
- ☐ D. 70 ÷ 0.39

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 4 yd? *1 pt*

12. 1 kg = 2.204 lb. How many pounds is 20 kg? *1 pt*

13. 1 gal = 3.785 L. How many liters is 5 gal? *1 pt*

14. 1 ft = 0.305 m. A patio is 20 ft^2 in area. Which is closest to its area in square meters? *1 pt*

- ○ A. 6.1 m^2
- ○ B. 0.093 m^2
- ○ C. 1.86 m^2
- ○ D. 12.2 m^2

15. Rate on Nov 5: 1 US dollar = 0.92 euros.

a. How many euros is $60? *1 pt*

b. How many US dollars is 23 euros? *1 pt*

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 26 miles per gallon and its tank holds 15 gal.

a. How far does it go on a full tank? *1 pt*

b. It makes a 1,000-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

17. A recipe for 4 servings uses $1\frac{1}{2}$ cups of rice.

a. How much rice for 10 servings? *1 pt*

b. With 6 cups on hand, how many servings can be made? *1 pt*

18. Juice is mixed orange : mango : water = 3 : 2 : 4. A batch uses 9 cups of orange. How many cups is the whole batch? *1 pt*

19. 1 mi = 1.61 km. A drive is 200 km long. Car A averages 40 mi/h; car B averages 60 km/h. Which car gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*